Bronchial reactivity and dietary antioxidants

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BACKGROUND: It has been postulated that dietary antioxidants may influence the expression of allergic diseases and asthma. To test this hypothesis a case-control study was performed, nested in a cross sectional study of a random sample of adults, to investigate the relationship between allergic disease and dietary antioxidants. METHODS: The study was performed in rural general practices in Grampian, Scotland. A validated dietary questionnaire was used to measure food intake of cases, defined, firstly, as people with seasonal allergic-type symptoms and, secondly, those with bronchial hyperreactivity confirmed by methacholine challenge, and of controls without allergic symptoms or bronchial reactivity. RESULTS: Cases with seasonal symptoms did not differ from controls except with respect to the presence of atopy and an increased risk of symptoms associated with the lowest intake of zinc. The lowest intakes of vitamin C and manganese were associated with more than fivefold increased risks of bronchial reactivity. Decreasing intakes of magnesium were also significantly associated with an increased risk of hyperreactivity. CONCLUSIONS: This study provides evidence that diet may have a modulatory effect on bronchial reactivity, and is consistent with the hypothesis that the observed reduction in antioxidant intake in the British diet over the last 25 years has been a factor in the increase in the prevalence of asthma over this period.